

STAT 579: Applied Causal Inference

Syllabus – Fall 2025

Instructor: Davood Tofighi, Ph.D.
Affiliation: Department of Mathematics and Statistics
Email: [\[Your UNM Email Address\]](#)
Office Hours: [Day and Time], [Location or Zoom Link]
Class Meetings: Tuesdays and Thursdays, 9:30 AM – 10:45 AM
Class Location: [Building and Room Number]

Course Overview

This graduate-level course provides a comprehensive introduction to the theory and application of causal inference, with a strong emphasis on biostatistical applications. We will move beyond association to learn formal frameworks for defining and estimating causal effects from observational data. The course covers foundational concepts like potential outcomes and causal graphs (DAGs), explores challenges such as confounding and selection bias, and details essential methods including propensity scores, instrumental variables, mediation analysis, and techniques for time-varying treatments.

My Approach to Causal Inference

This course emphasizes that causal inference is a way of thinking rigorously about cause and effect. We will constantly return to foundational principles: clearly defining the causal question, making all assumptions explicit (often with DAGs), understanding what each method can and cannot do, and critically assessing the robustness of our conclusions.

Learning Objectives

Upon successful completion of this course, students will be able to:

1. Distinguish between association and causation and formally define causal effects using the potential outcomes framework.
2. Represent causal structures using Directed Acyclic Graphs (DAGs) and use them to identify confounding and selection bias.
3. Articulate and critically evaluate the assumptions required for causal identification (e.g., exchangeability, positivity, SUTVA, exclusion restriction).
4. Apply propensity score methods (matching, weighting) to estimate causal effects, including checking diagnostics like balance and overlap.

5. Implement g-computation and inverse probability weighting for estimating causal effects of time-varying treatments using Marginal Structural Models.
6. Decompose effects into natural direct and indirect effects using causal mediation analysis, understanding the strong assumptions required.
7. Conduct sensitivity analyses to evaluate the robustness of causal estimates to violations of key assumptions.

Assignments and Grading Policies

Assessment Components

Your final grade is a reflection of your understanding of theoretical concepts, your ability to apply methods, and your engagement with the material.

Weekly Problem Sets (20%): Short assignments focusing on theoretical concepts, DAG interpretation, and articulating assumptions.

Applied R Assignments (25%): A series of five assignments where you will independently apply methods to real or synthetic datasets.

Midterm Exam (20%): Covers foundational concepts from the first half of the course (Weeks 1-6).

Final Project (35%): A capstone project applying causal methods to a dataset. The project includes:

Project Proposal (5% of total grade): A clear causal question, dataset identification, and an initial DAG. Due mid-semester.

Final Report & Presentation (30% of total grade): A comprehensive report detailing your methods, findings, and limitations, accompanied by a short in-class presentation. Due during finals week.

Grading Scale

Final letter grades will be assigned based on the following scale:

A+	97-100%	B-	80-82.9%
A	93-96.9%	C+	77-79.9%
A-	90-92.9%	C	73-76.9%
B+	87-89.9%	C-	70-72.9%
B	83-86.9%	F	< 70%

Recommended Materials

Primary Texts: • Hernán MA, Robins JM. *Causal Inference: What If*. ([Online](#))

- Pearl J, et al. *Causal Inference in Statistics: A Primer*.
- Imbens GW, Rubin DB. *Causal Inference for Statistics, Social, and Biomedical Sciences*.

Supplementary Texts: • VanderWeele TJ. *Explanation in Causal Inference*.

- Brumback BA. *Fundamentals of Causal Inference with R*.

Software: We will use **R** and **RStudio**. Key packages will include **dagitty**, **MatchIt**, **WeightIt**, **cobalt**, **ivreg**, and **mediation**.

Course Schedule

1	Introduction		Association vs. Causation, Simpson's Paradox, Intro to Potential Outcomes & <i>Readings</i> : HR Ch 1; PGJ Ch 1
2	Potential Outcomes		ATE/ATT, SUTVA, Consistency, Randomized Experiments & <i>Readings</i> : HR Ch 2-3; IR Ch 2-3
3	Causal Graphs (DAGs)		Nodes, Edges, Paths, d-separation, Backdoor Criterion & <i>Readings</i> : PGJ Ch 2; HR Ch 6-7
4	Bias and Identification		Colliders, M-Bias, Selection Bias, Frontdoor Criterion & <i>Readings</i> : PGJ Ch 3, HR Ch 8
5	G-Formula		Conditional Exchangeability, Positivity, Standardization & <i>Readings</i> : HR Ch 4-5
6	Propensity Scores		Balancing Scores, Matching, Stratification, IPTW & <i>Readings</i> : HR Ch 12-13; IR Ch 13-18
7	Midterm Exam		Covers material from Weeks 1-6.
8	Instrumental Variables		Exclusion Restriction, LATE, 2SLS. Project Proposal Due. & <i>Readings</i> : HR Ch 16; IR Ch 23
9	Causal Mediation I		NDE, NIE, Counterfactual Definitions, Identification Assumptions & <i>Readings</i> : VanderWeele Ch 1-2
10	Causal Mediation II		Estimation (Difference/Product Methods), Interaction & <i>Readings</i> : VanderWeele Ch 3-4
11	Sensitivity Analysis		E-values, Bounding Factors, Robustness to Unmeasured Confounding & <i>Readings</i> : HR Ch 14; Ding & VanderWeele (2016)
12	Marginal Structural Models		Time-Varying Confounding, IPW for MSMs, Sequential Exchangeability & <i>Readings</i> : HR Ch 17-18, 21
13	Project Work		In-class workshop, final project support, and Q&A.

Course Policies

Late Work Policy: Deadlines are firm. Late assignments will be penalized 10% for each day they are late. Exceptions will only be granted for documented emergencies, and you must contact

the instructor as soon as possible, preferably before the deadline.

Academic Integrity: All work submitted in this course must be your own. Collaboration on high-level concepts for problem sets is encouraged, but all submitted code and written answers must be your own original work. Copying code from classmates or online sources without attribution is a serious violation of academic integrity. Please consult the university's academic integrity policy for detailed information.

Communication: The primary method for course announcements will be via the university's learning management system. For questions about course content, please use office hours or ask in class. For personal or administrative matters, email the instructor; please allow 24-48 business hours for a response.

Classroom Environment: This course is designed to be an inclusive and welcoming environment for all students. Diverse viewpoints are valued and encouraged. All students are expected to contribute to a respectful and collaborative atmosphere during class discussions.

Accessibility: Students with disabilities who may need accommodations in this class are encouraged to contact the university's Accessibility Resource Center as soon as possible to ensure that such accommodations are arranged in a timely fashion.